

Ex.No :7

Use AF Logical OSE to extract data from an Android device

Date :

Description

AFLogical OSE (Open Source Edition) is a tool used to extract data from Android devices. It's part of the Open Source Android Forensics project and is often used for logical extraction, which means pulling data like contacts, messages, and call logs without directly accessing the file system. Here's a step-by-step guide to using AFLogical OSE to extract data from an Android device.

Step 1: Prepare Your Environment

1. Download AFLogical OSE:

- Visit the [AFLogical OSE GitHub page](#) and download the latest release. Alternatively, you can clone the repository using Git.

2. Install Java (if not already installed):

- AFLogical OSE requires Java to run. Make sure you have Java installed on your machine. You can download it from the [official Java website](#).

3. Install Android Debug Bridge (ADB):

- ADB is a command-line tool that allows communication with Android devices. Download it from the [Android Developer's website](#) and install it on your system.
- Add ADB to your system's PATH environment variable to easily run ADB commands from the terminal or command prompt.

4. Enable USB Debugging on the Android Device:

- On the Android device, go to Settings > About Phone and tap Build Number seven times to enable Developer Options.
- Go to Settings > Developer Options and enable USB Debugging.

Step 2: Connect the Android Device to Your Computer

1. Connect via USB:

- Use a USB cable to connect the Android device to your computer.

2. Verify ADB Connection:

- Open Command Prompt (Windows) or Terminal (macOS/Linux) and run:

```
bash
```

```
adb devices
```

- This command lists connected devices. If your device is listed, you are ready to proceed. If not, ensure that USB Debugging is enabled and that the correct drivers are installed.

Step 3: Extract Data Using AFLogical OSE

1. Run AFLogical OSE:

- Navigate to the directory where you have downloaded or extracted AFLogical OSE.

2. Push the APK to the Android Device:

- Run the following command to install the AFLogical OSE APK on the device:

bash

```
adb install aflogical.apk
```

- This installs the AFLogical app on the device, which will be used to collect data.

3. Launch the AFLogical OSE App on the Device:

- On the Android device, open the AFLogical app.
- You'll see options to extract various types of data (e.g., contacts, SMS, MMS, call logs). Select the data types you want to extract.

4. Start Data Extraction:

- Once you've selected the data types, the app will start the extraction process.
- Data is stored in .csv files on the device's storage, typically in a directory named aflogical.

Step 4: Transfer Extracted Data to Your Computer

1. Use ADB to Pull Data:

- After extraction, you need to transfer the data files from the Android device to your computer. Run:

bash

```
adb pull /sdcard/aflogical /path/to/destination
```

- Replace /path/to/destination with the directory path where you want to save the extracted data on your computer.

2. Verify the Data:

- Navigate to the destination directory and check the .csv files to ensure all required data has been extracted.

Step 5: Analyze the Extracted Data

1. Open the CSV Files:

- Use Excel, Google Sheets, or any text editor to open and analyze the .csv files. These files contain the extracted data like contacts, SMS, MMS, and call logs.

2. Review and Document:

- Carefully review the data for any relevant information. Document your findings and consider exporting the data into a report format for further analysis.

Step 6: Clean Up

1. Uninstall AFLogical OSE:

- Once data extraction is complete, you can uninstall the AFLogical OSE app from the Android device by running:

bash

```
adb uninstall com.viaforensics.android.aflogical
```

2. Disconnect the Device:

- Safely disconnect the Android device from your computer.

By following these steps, you can use AFLogical OSE to effectively extract and analyze data from an Android device. This process is valuable for forensic investigations, especially when dealing with logical data that doesn't require root access to the device.

OUTPUT :

Installing JAVA :

```
C:\Users\nithin>java -version
openjdk version "25.0.4.1" 2026-08-18 LTS
OpenJDK Runtime Environment Temurin-25.0.4.1+1 (build 25.0.4.1+1-LTS)
OpenJDK 64-Bit Server VM Temurin-25.0.4.1+1 (build 25.0.4.1+1-LTS, mixed mode, sharing)

C:\Users\nithin>
```

Installing ADB Tools :

```
C:\Users\nithin\Downloads\platform-tools-latest-windows\platform-tools>adb version
Android Debug Bridge Version 1.0.41
Version 37.0.1-15733141
Installed as C:\Users\nithin\Downloads\platform-tools-latest-windows\platform-tools\adb.exe
Running on Windows 10.0.26200

C:\Users\nithin\Downloads\platform-tools-latest-windows\platform-tools>
```

ADB Device Connection :

```
C:\Users\nithin\Downloads\platform-tools-latest-windows\platform-tools>adb devices
* daemon not running; starting now at tcp:5037
* daemon started successfully
List of devices attached
ZD2227X3N4      device

C:\Users\nithin\Downloads\platform-tools-latest-windows\platform-tools>
```

Installing aflogical.apk :

```
C:\Users\nithin\Downloads\platform-tools-latest-windows\platform-tools>adb install
com.viaforensics.android.aflogical_ose_1.5.2_OSE_152.apk
Performing Streamed Install
Success

C:\Users\nithin\Downloads\platform-tools-latest-windows\platform-tools>
```

Capturing SMS and Call Logs :

```
C:\Users\nithin\Downloads\platform-tools-latest-windows\platform-tools>.adb.exe shell content query --uri
content://sms > sms.txt

C:\Users\nithin\Downloads\platform-tools-latest-windows\platform-tools>.adb.exe shell content query --uri
content://call_log/calls > call_logs.txt

C:\Users\nithin\Downloads\platform-tools-latest-windows\platform-tools>
```

Extracted SMS Logs :

```
Row: 0 _id=6810, thread_id=2294, address=919032675630, person=NULL, date=1788759138655, date_sent=1788759138000, protocol=0, read=0, status=-1, type=1, reply_path_present=0,
subject=proto:CjokImVbS5nb29nbGUuYm5kcm9pZC5hcHBzLm1lc3NhZ2luZy4SFClAKhBR9rhBzs1NWKcgbNwsYPLM, body=Dear Customer, You have a missed call from +919032675630 The last missed call was at
10:58 AM on 07-Sep-2026 Thankyou, Team Jio., service_center=+919014300414, locked=0, sub_id=2, error_code=-1, creator=com.google.android.apps.messaging, seen=1
Row: 1 _id=6809, thread_id=1807, address=JK-NSESMS-S, person=NULL, date=1788644789072, date_sent=1788644785000, protocol=0, read=0, status=-1, type=1, reply_path_present=0,
subject=proto:CjokImVbS5nb29nbGUuYm5kcm9pZC5hcHBzLm1lc3NhZ2luZy4SFClAKhBcdOffIPtNS5jVNRDAlwIM, body=GROWMINVESTTECHPRIVATELIMITED on 29-08-2026 reported your Fund bal Rs.0.000 &
Securities bal 0.000. This excludes your Bank, DP & PMS bal with the broker-NSE, service_center=+917003075026, locked=0, sub_id=2, error_code=-1,
creator=com.google.android.apps.messaging, seen=1
Row: 2 _id=6808, thread_id=2237, address=JK-IRMSA-S, person=NULL, date=1788533923394, date_sent=1788533919000, protocol=0, read=0, status=-1, type=1, reply_path_present=0,
subject=proto:CjokImVbS5nb29nbGUuYm5kcm9pZC5hcHBzLm1lc3NhZ2luZy4SFClAKhC4+HvgX/VE169aMKRrZ3AD, body=PNR-4444128174
Trn:22715
Dt:05-09-26 Dep.Time:09:47 Hrs.
Frm KRNT to MDU
Cls:SL
P1-S4,30
P2-S4,29
P3-S4,32
Boarding allowed from KRNT only
Chart Prepared
Url for coach position: https://enquiry.indianrail.gov.in/mntes/c?u=ggIfjNa9d7NejengkN8I
Download RailOne for latest updates https://railone.indianrailways.gov.in
For Enquiry/Complaint/Assistance, please dial 139 IR-CRIS, service_center=+917019075011, locked=0, sub_id=2, error_code=-1, creator=com.google.android.apps.messaging, seen=1
Row: 3 _id=6807, thread_id=1771, address=JM-NSESMS-S, person=NULL, date=1788530637133, date_sent=1788530635000, protocol=0, read=0, status=-1, type=1, reply_path_present=0,
subject=proto:CjokImVbS5nb29nbGUuYm5kcm9pZC5hcHBzLm1lc3NhZ2luZy4SFClAKhD1EKi8KqpfLZ4Q+yhN1/5b, body=Beware while dealing based on unsolicited tips through
Whatsapp,telegram,SMS,calls,Youtube,etc,& take an informed decision. Visit https://bit.ly/3QqX5Co -NSE, service_center=+917021075016, locked=0, sub_id=2, error_code=-1,
creator=com.google.android.apps.messaging, seen=1
Row: 4 _id=6806, thread_id=1806, address=JA-JIOPAY-S, person=NULL, date=1788522742702, date_sent=1788522691000, protocol=0, read=0, status=-1, type=1, reply_path_present=0,
subject=proto:CjokImVbS5nb29nbGUuYm5kcm9pZC5hcHBzLm1lc3NhZ2luZy4SFClAKhDSw0aszQ9GRYpwins76spK, body=హలో!
మీరు 04-Sep-26 17:21 వరకు మీ రోజువారీ డేటాలో 100% వినియోగించారు.
Jio నెంబర్: 8074999510
రోజువారీ డేటా కోటా: 2 GB
రోజువారీ కోటా 05-Sep-26 00:27 న మళ్ళీ అందుబాటులో ఉంటుంది.
మీ వద్ద అదనపు డేటా ప్యాక్ లేకపోతే, మీ ఇంటర్నెట్ వేగం 64 Kbps వరకు తగ్గుతుంది.
MyJio యాప్ ద్వారా డేటా ప్యాక్ రీడార్డ్ చేయడానికి క్లిక్ చేయండి: http://tiny.jio.com/dbuydatapack
```

Extracted Call Logs :

```
Row: 0 formatted_number=NULL, duration=59, subject=NULL, is_call_log_phone_account_migration_pending=0, subscription_id=3, photo_id=0, post_dial_digits=, call_screening_app_name=NULL, priority=0, number=, countryiso=IN, cnap_name=, photo_uri=NULL, geocoded_location=India, missed_reason=0, block_reason=0, cached_convert_number=NULL, subscription_component_name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, add_for_all_users=1, numbertype=NULL, features=68, transcription=NULL, last_modified=1729700812435, _id=8385, new=0, date=1718092698184, name=Seenu, type=1, presentation=1, via_number=, numberlabel=NULL, normalized_number=, composer_photo_uri=NULL, phone_account_address=, phone_account_hidden=0, lookup_uri=NULL, voicemail_uri=NULL, matched_number=NULL, transcription_state=0, data_usage=NULL, location=NULL, call_screening_component_name=NULL, is_read=1
Row: 1 formatted_number=NULL, duration=0, subject=NULL, is_call_log_phone_account_migration_pending=0, subscription_id=NULL, photo_id=0, post_dial_digits=, call_screening_app_name=NULL, priority=0, number=, countryiso=IN, cnap_name=NULL, photo_uri=NULL, geocoded_location=India, missed_reason=0, block_reason=0, cached_convert_number=NULL, subscription_component_name=NULL, add_for_all_users=1, numbertype=NULL, features=0, transcription=NULL, last_modified=1729700812435, _id=8417, new=0, date=1718254552633, name=, type=2, presentation=1, via_number=, numberlabel=NULL, normalized_number=, composer_photo_uri=NULL, phone_account_address=NULL, phone_account_hidden=0, lookup_uri=NULL, voicemail_uri=NULL, matched_number=NULL, transcription_state=0, data_usage=NULL, location=NULL, call_screening_component_name=NULL, is_read=1
Row: 2 formatted_number=+91 90520 33059, duration=0, subject=NULL, is_call_log_phone_account_migration_pending=0, subscription_id=3, photo_id=0, post_dial_digits=, call_screening_app_name=NULL, priority=0, number=, countryiso=IN, cnap_name=, photo_uri=, geocoded_location=India, missed_reason=1114112, block_reason=0, cached_convert_number=NULL, subscription_component_name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, add_for_all_users=1, numbertype=0, features=64, transcription=NULL, last_modified=1729700812435, _id=8461, new=0, date=1718982030862, name=Sami, type=3, presentation=1, via_number=, numberlabel=Mobile, normalized_number=, composer_photo_uri=NULL, phone_account_address=, phone_account_hidden=0, lookup_uri=content://com.android.contacts/contacts/lookup/1881373dd0188e90969c/4201?directory=0, voicemail_uri=NULL, matched_number=NULL, transcription_state=0, data_usage=NULL, location=NULL, call_screening_component_name=NULL, is_read=1
Row: 3 formatted_number=+91 77991 68016, duration=38, subject=NULL, is_call_log_phone_account_migration_pending=0, subscription_id=3, photo_id=0, post_dial_digits=, call_screening_app_name=NULL, priority=0, number=+917799168016, countryiso=IN, cnap_name=+917799168016, photo_uri=, geocoded_location=India, missed_reason=0, block_reason=0, cached_convert_number=NULL, subscription_component_name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, add_for_all_users=1, numbertype=0, features=64, transcription=NULL, last_modified=1729700812435, _id=8499, new=0, date=1720318909704, name=, type=1, presentation=1, via_number=, numberlabel=Mobile, normalized_number=+917799168016,
```

Extracted Device Properties :

```
PS C:\Users\nithin\Downloads\platform-tools-latest-windows\platform-tools> Write-Host "Device Model:" (.adb.exe shell getprop ro.product.model)
Device Model: One Plus Nord CE4

PS C:\Users\nithin\Downloads\platform-tools-latest-windows\platform-tools> Write-Host "Android Version:" (.adb.exe shell getprop ro.build.version.release)
Android Version: 16

PS C:\Users\nithin\Downloads\platform-tools-latest-windows\platform-tools> Write-Host "Security Patch Level:" (.adb.exe shell getprop ro.build.version.security_patch)
Security Patch Level: 2026-07-01

PS C:\Users\nithin\Downloads\platform-tools-latest-windows\platform-tools>
```

Uninstalling aflogical.apk :

```
C:\Users\nithin\Downloads\platform-tools-latest-windows\platform-tools>.adb.exe uninstall --user 0
com.viaforensics.android.aflogical_ose_1.5.2_OSE_152.apk
```

```
C:\Users\nithin\Downloads\platform-tools-latest-windows\platform-tools>
```

Rubrics

Criteria&Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity&Submission Regularity	3	
2. Application of Forensic Tools&Practical Execution	3	
3. Documentation&Reporting	2	
4. Engagement, Problem-Solving&Team Collaboration	2	
Total	10	

Result :

The experiment successfully performed a logical extraction of data from the Android device using the A-FLogical OSE framework via ADB interface connection. The tool bypassed file system access limitations to cleanly pull device properties, **SMS logs, and call records into structured .csv files**, which were verified on the local host before safely uninstalling the APK tool.